Comprehensive Research Dossier: The Life, Laboratory, and Legacy of Kary Banks Mullis (1944–2019)

This research report serves as an exhaustive collection of primary and high-signal secondary materials documenting the professional and personal history of Kary Mullis. It is designed to support the construction of a narrative profile for a BIOL 3030 introductory biochemistry audience, emphasizing the discovery of the Polymerase Chain Reaction (PCR) and the human-technical interface of 1980s biotechnology.

A) ANCHOR SCENES

SCENE #1: The Night on Highway 128: The Geometry of DNA Doubling

Summary: While driving a silver Honda Civic from the San Francisco Bay Area to a weekend cabin in Mendocino County, Kary Mullis conceptualizes the mathematical and biochemical framework for the exponential amplification of DNA. This event is the definitive "Eureka" moment in the history of PCR, where the recursive logic of using two opposing primers to create a self-catalytic doubling cycle first occurs to him.

Date/Place: A Friday night in the late spring of 1983 (approximately April or May); Highway 128, specifically between Cloverdale and Boonville, California, around mileage marker 46.58.

People: Kary Mullis (driver); Jennifer Barnett (passenger, asleep; fellow chemist at Cetus Corporation).

Concrete details: The setting was characterized by a "weird" atmosphere, saturated with the scent of California buckeyes. Mullis describes "lurid blue and pink images of electric molecules" injecting themselves into his vision between the road and his eyes. The technical trigger was a thought experiment aimed at solving the problem of identifying single nucleotide polymorphisms (SNPs). He was mentally wrestling with the "sham reaction" method—a way to use up stray deoxynucleoside triphosphates (dNTPs) in a sample before adding radioactive traces.

The epiphany arrived in stages. First, he realized that using two primers—one for the upper strand and one for the lower—would create a specific DNA fragment of a defined length. Then, the mathematical reality of recursive doubling hit him. He stopped the car twice, once on the road's edge and again at a pullout, to perform mental arithmetic. He calculated that ten cycles would yield a thousand-fold increase, twenty cycles a million-fold, and thirty cycles a billion-fold ($2^{30} \approx 10^9$), which roughly equaled the number of base pairs in the human genome. Despite his excitement, Jennifer Barnett remained asleep throughout the

calculations; Mullis reportedly shouted that this discovery would make him famous and win him the Nobel Prize.

Why it matters: This scene represents the shift from "linear" thinking to "exponential" thinking in molecular biology. Before PCR, DNA analysis was restricted by the "fuzziness" of genomic DNA and the slow, inevitable delays of cloning. Mullis's realization bypassed the need for living bacterial hosts to replicate DNA, effectively "digitizing" DNA replication into a test-tube environment.

Evidence quality: Primary source documentation is exceptionally strong but largely subjective, based on Mullis's Nobel lecture and his memoir *Dancing Naked in the Mind Field*. While the technical details are consistent, the sensory details are retrospective.

Myth risk: High. The "Eureka" drive is a core component of Mullis's personal brand. While the location and technical leap are documented, the "perfect" timing of the mental leaps and the dramatic silence of the sleeping passenger are classic narrative tropes that may be stylized.

Sources:

SCENE #2: PCR01 and the "Purple-Capped Tube" Failure

Summary: Mullis conducts the first physical attempt at the polymerase chain reaction in a laboratory setting. This experiment, labeled "PCR01," was a rudimentary test of the concept using a single tube and isothermal conditions. It failed to produce the expected results but marked the transition from mental architecture to empirical trial.

Date/Place: September 8, 1983; Cetus Corporation, Emeryville, California.

Who was involved: Kary Mullis (lead investigator); Fred Faloona (research assistant).

Concrete texture: The experiment was conducted in a "single, purple-capped tube". The methodology was notably "isothermal"—Mullis hoped the reaction might "cycle itself" due to the natural chemical equilibrium between double-stranded and single-stranded DNA. He cut the template DNA with a restriction enzyme and added a "sumptuous abundance" of primers. The post-reaction detection involved ethidium bromide-stained agarose gel electrophoresis. The resulting image was a large, indistinct "smear" rather than a discrete band, which was met with profound skepticism by other Cetus scientists.

Why it matters: This scene highlights the disconnect between a theoretical "beautiful idea" and the physical constraints of enzymology. It demonstrates Mullis's "chemical prejudice"—the belief that thermodynamic equilibrium would favor the reaction without mechanical intervention. The failure of PCR01 forced the team to accept that discrete temperature steps—specifically a

high-temperature denaturation step—were non-negotiable, leading to the manual (and eventually automated) thermal cycling process.

Evidence quality: Strong. This is supported by specific lab notebook entries (p. 6, PCR01) and photographs of the original notebook, which were later published.

Myth risk:

Low. The failure of the first experiment is a standard part of the scientific record and lacks the romanticism of the Highway 128 drive.

Sources:

SCENE #3: The "Interpersonal Wrecking Ball" at Cetus

Summary: As PCR moves from an idea to a development project, Mullis's volatile personality creates severe institutional friction. While his friend Tom White protects his employment, the "top-notch experimentalists" at Cetus struggle to replicate his work and resent his lack of rigorous controls.

Date/Place: Summer 1984; Cetus Corporation, Emeryville, California.

People: Kary Mullis; Tom White (VP of Research); Henry Erlich (Director of Human Genetics); Randy Saiki (Researcher); Norman Arnheim (Researcher).

Concrete details: Mullis was described as an "interpersonal wrecking ball" who fought with everyone from senior management to security guards. He engaged in public "lovers' quarrels" with a fellow chemist at the firm and reportedly threatened to bring a gun to the office. At a staff party, he "nearly came to blows" with another scientist. Because Mullis produced ambiguous results and resisted running "appropriate controls and repetition," his colleagues remained circumspect about whether PCR actually worked on complex genomic DNA. Tom White eventually reassigned Mullis to work on PCR full-time, while placing more disciplined scientists like Randy Saiki on parallel projects to validate the technique.

Why it matters: This scene demonstrates that the "invention" of a technology is not merely a mental act but an institutional one. Without the rigorous validation provided by Saiki and Erlich—and the political protection of White—PCR might have been abandoned as a flawed theory by an unreliable employee. It highlights the "collaboration through conflict" that characterized Cetus's culture.

Evidence quality: High. Well-documented through oral histories with Cetus staff and articles in *California Magazine*.

Myth risk:

Medium. The "mad scientist" trope is prevalent, but the specific details (gun threats, party fights) are consistently reported by multiple witnesses.

Sources:

SCENE #4: The Cold Spring Harbor "Genome Shock"

Summary: Mullis presents PCR to the global scientific elite at the 1986 Cold Spring Harbor Symposium. Despite the revolutionary nature of the technique, the presentation is initially overshadowed by the massive financial and ethical debate surrounding the nascent Human Genome Project (HGP).

Date/Place: May 1986; Cold Spring Harbor Laboratory, New York.

People: Kary Mullis (presenter); James Watson (organizer); Wally Gilbert (Nobel laureate); Tom Caskey (session summarizer).

Concrete details: The meeting took place in the Grace Auditorium. Mullis described the PCR process, yet Tom Caskey, in his official summary of the event, failed to mention PCR at all. The audience was instead preoccupied with Wally Gilbert's announcement that sequencing the human genome would cost $3.5 billion, a figure that caused "shock" and fears that it would drain money from investigator-initiated research. Mullis had recently been rejected by *Nature* for his PCR manuscript, adding to his sense of alienation from the "high priests" of biology.

Why it matters: This scene illustrates the "lag" in scientific recognition. Even in an audience of experts, the significance of a tool that would eventually make the Human Genome Project feasible was not immediately grasped. It also fueled Mullis's lifelong disdain for peer-review and scientific gatekeepers.

Evidence quality: Strong. Official records of the 1986 symposium and eyewitness accounts by James Watson and others provide a detailed context.

Myth risk: Low. The lack of immediate recognition for PCR is a documented irony of the era.

Sources:

SCENE #5: The DuPont Trial: Defining "Invention"

Summary: Mullis testifies in a high-stakes legal battle between Cetus and DuPont. DuPont argued that Mullis did not "invent" PCR because a 1971 paper by Kjell Kleppe and H. Gobind Khorana had already described the principles of using two primers and a polymerase. Mullis had to defend the "non-obviousness" and practicality of his invention in a court of law.

Date/Place: February 1991; U.S. District Court, San Francisco.

People: Kary Mullis (witness); Arthur Kornberg (Nobel laureate, witness for DuPont); Tom White (witness for Cetus).

Concrete details: DuPont produced the 1971 Kleppe paper as "prior art". Mullis testified about a 1984 encounter where he showed a PCR poster to Nobelist Joshua Lederberg, who admitted he and Kornberg had "considered the notion" but failed to realize the "chain reaction" logic because oligonucleotides were then unavailable. The legal defense rested on the distinction between a theoretical chemical reaction and a "practical laboratory method". The jury eventually ruled in favor of Cetus, confirming Mullis as the sole legal inventor of PCR.

Why it matters: The trial was the ultimate crucible for Mullis's legacy. It defined the boundaries between "discovery" and "invention" and paved the way for the $300 million sale of the PCR patents to Roche, which Mullis later felt cheated by.

Evidence quality: Excellent. Based on trial transcripts and depositions preserved in the Center for Public Genomics.

Myth risk:

Low. Legal proceedings are among the best-documented aspects of Mullis's career.

Sources:

B) DIRECT QUOTES FROM THE SCIENTIST

"DNA before PCR was long and stringy, not really molecular at all."

- **Context:** 1993 Nobel Lecture, Stockholm.
- **Source:** Mullis, K. B. (1993, December 8). The Polymerase Chain Reaction. Nobel Lecture..

"I think Feynman would have said, if you can understand it, you can explain it. So I'm going to try to explain how it was that I invented the polymerase chain reaction."

- **Context:** 1993 Nobel Lecture, Stockholm.
- **Source:** Mullis, K. B. (1993, December 8). The Polymerase Chain Reaction. Nobel Lecture..

"Dear Thor! I exclaimed. With two oligonucleotides, DNA polymerase, and the four nucleoside triphosphates I could make as much of a DNA sequence as I wanted and I could make it on a fragment of a specific size that I could distinguish easily."

- **Context:** Describing the Highway 128 epiphany in his 1990 *Scientific American* article.

- **Source:** Mullis, K. B. (1990). The unusual origin of the polymerase chain reaction. *Scientific American*, 262(4), 56-61..

"EUREKA!!!! The result would be exactly the same only the signal strength would be doubled."

- **Context:** Explaining the realization of the doubling effect on the drive.
- **Source:** Mullis, K. B. (1993, December 8). The Polymerase Chain Reaction. Nobel Lecture..

"Science, like nothing else among the institutions of mankind, grows like a weed every year. Art is subject to arbitrary fashion, religion is inwardly focused and driven only to sustain itself, law shuttles between freeing us and enslaving us."

- **Context:** His personal philosophy on the growth of scientific knowledge.
- **Source:** Mullis, K. B. (1998). *Dancing Naked in the Mind Field*. Pantheon Books..

"It was not easy in that post-cloning, pre-PCR year to accept the fact that you could have all the DNA you wanted. And that it would be easy."

- **Context:** Reflecting on the skepticism of the molecular biology community.
- **Source:** Mullis, K. B. (1993, December 8). The Polymerase Chain Reaction. Nobel Lecture..

C) HUMAN REALITY DETAILS

- Mullis grew up in rural North Carolina and South Carolina, where he and his cousins would taunt livestock by feeding them through electric fences.
- As a teenager, he and his friends built 4-foot-long solid-fuel rockets using potassium nitrate and sugar melted on a charcoal stove, once launching a frog a mile into the air and recovering it alive.
- In high school, he had "unsupervised access" to the chemistry lab, a freedom he later claimed would make him a "menace to society" by modern standards.
- While an undergraduate at Georgia Tech, he and Al Montgomery built an organic synthesis lab in an old chicken house to sell "noxious or either explosive" research chemicals.
- His doctoral dissertation at Berkeley was accepted only after friends "cut all the whacko stuff out of it" and his advisor lobbied the committee to overlook his lack of general biochemistry knowledge.
- He published a sole-author paper in *Nature* in 1968 titled "The Cosmological Significance of Time Reversal," which he later admitted was a "sophomoric astrophysical hypothesis".

- He became a "Universal Life minister" during his "hippie days" at Berkeley and performed the wedding ceremony for his friend Tom White.
- Between his postdoctoral fellowships and joining Cetus, Mullis quit science briefly to write fiction and managed a bakery/cafe owned by his first wife.
- He admitted he "really didn't know much of anything about molecular biology" when he was hired by Cetus to synthesize oligonucleotides.
- Mullis was notorious at Cetus for his "interpersonal wrecking ball" behavior, including threatening to bring a gun to work and having public lovers' quarrels.
- He was a regular user of LSD and believed the drug helped him visualize the recursive logic of PCR.
- He claimed to have encountered a "fluorescent raccoon" at his cabin in 1985, which he speculated might be an extraterrestrial.
- In 1992, he founded "Star Gene" to sell jewelry containing the amplified DNA of dead celebrities like Marilyn Monroe.
- Minutes before his Nobel lecture, he reportedly told an organizer he wanted to talk about why HIV does not cause AIDS rather than PCR.
- Mullis's favorite scientific retreating place was a cabin in the mountains of Mendocino, where he worked on his personal curiosity away from "bosses".

D) SCIENCE CONNECTION (INTRO BIOCHEMISTRY)

Best-fit Topic

DNA Metabolism and Analytical Methods (Nucleic Acid Amplification).

Precise Contribution

Kary Mullis conceptualized and developed the Polymerase Chain Reaction (PCR), a site-specific, recursive in vitro method for the exponential amplification of DNA sequences using thermal cycling and oligonucleotide primers.

Bridge Explanation

PCR mimics natural DNA replication but uses thermal energy instead of enzymes like helicase to separate the double helix (denaturation). By introducing specific primers that flank a target sequence, scientists can dictate where DNA polymerase begins copying. This process creates a "chain reaction" where the product of one cycle becomes the template for the next, doubling the target concentration every few minutes.

Scientist's Analogy

Mullis compared the challenge of finding a specific gene in the human genome to "reading a particular license plate out on Interstate 5 at night from the moon." He explained that while scientists had powerful enough "telescopes" (probes), they couldn't see the plate because it was

surrounded by millions of other cars; PCR works by making millions of copies of just that one car so it becomes the only thing visible.

Key Equations in PCR

The theoretical yield of DNA follows the exponential growth formula:

$$Y = X \cdot 2^n$$
Where:

- Y = final number of DNA copies
- X = initial number of template molecules
- n = number of thermal cycles.

E) COMPLICATIONS OR CONTROVERSIES

1. The Priority Dispute (Kleppe/Khorana 1971)

- **What's agreed:** In 1971, Kjell Kleppe and H. Gobind Khorana published a paper describing the use of two primers and DNA polymerase to replicate a tRNA gene.
- **What's disputed:** DuPont argued this made Mullis's invention "obvious." Mullis and Cetus argued Kleppe's method was neither exponential nor practical because automated oligonucleotide synthesis did not exist yet.
- **Sources:**

2. Credit Assignment (Mullis vs. the Cetus Team)

- **What's agreed:** Mullis had the "Eureka" idea, but the first successful amplification of a human gene (beta-globin) was performed by Randy Saiki, Henry Erlich, and others.
- **What's disputed:** Whether Mullis deserved a solo Nobel Prize. Colleagues like Erlich and Saiki felt their experimental validation was the actual breakthrough, as Mullis's initial results were ambiguous.
- **Sources:**

3. "Nobel Disease" (AIDS and Climate Denial)

- **What's agreed:** Mullis publicly stated that humans do not cause climate change and that HIV is not the cause of AIDS.
- **What's disputed:** Mullis claimed his skepticism was scientific, citing a lack of "primary literature" for the HIV-AIDS link. Biologists viewed this as dangerous bad-faith questioning.
- **Sources:**

F) "NOT ON WIKIPEDIA" FINDS

- **The Zinc-Free Buffer Discovery:** Mullis realized Bacterial Alkaline Phosphatase (BAP) could be heat-denatured if the buffer lacked zinc. This obscure chemical realization was a key step in his mental troubleshooting on the night of the PCR discovery.
- **The "Lederberg Head" Testimony:** During the 1991 trial, Mullis recounted how Joshua Lederberg turned his "enormous Nobel-laureated head" and admitted he and Kornberg had missed the chain reaction logic because they weren't thinking about oligonucleotides.
- **The "Purple Cap" Origin:** The first PCR experiment (PCR01) was specifically conducted in a tube with a purple cap, a detail Mullis preserved in his notebook to distinguish it from standard clear tubes.
- **The Coffee Shop Gap:** Mullis was so disillusioned with science after his postdocs that he managed a cafe for two years; he was only lured back by Tom White, who found him there and convinced him to join Cetus.
- **The "Menace to Society" Quote:** Mullis explicitly used this phrase in his Nobel lecture to describe his high school years, providing a rare primary-source critique of modern lab safety culture.

G) SOURCES CONSULTED

- **Mullis, K. B. (1993, December 8).** *The Polymerase Chain Reaction.* Nobel Lecture.
- **Mullis, K. B. (1990).** *The unusual origin of the polymerase chain reaction.* Scientific American, 262(4), 56-65.
- **Mullis, K. B. (1998).** *Dancing Naked in the Mind Field.* Pantheon Books.
- **Saiki, R. K., et al. (1985).** *Enzymatic amplification of beta-globin genomic sequences and restriction site analysis for diagnosis of sickle cell anemia.* Science, 230(4732), 1350–1354.
- **Center for Public Genomics.** *E.I. du Pont de Nemours & Co. v. Cetus Corporation.* Duke University Archives.
- **Cold Spring Harbor Laboratory.** *1986 Symposium Records.*
- **California Magazine (Winter 2019).** *Intolerable Genius: Berkeley's Most Controversial Nobel Laureate.*
- **Science History Institute.** *The Nobel Disease.*
- **BioFire Diagnostics.** *Remembering Kary Mullis.*
- **Nature (1968).** *The Cosmological Significance of Time Reversal.*

H) FLAGGED "MYTH-RISK" DETAILS

- **Claim:** The discovery happened exactly at mileage marker 46.58.
    - **Where it appears:** Mullis's autobiography and Nobel lecture.
    - **Why sourcing is weak:** This precision is unlikely in a moving car at night and serves a narrative function to ground a "mystical" experience in physical reality.
    - **Best evidence:** Mullis’s own subjective recollection is the only source.
- **Claim:** Mullis knew instantly he would win the Nobel Prize.

- **Where it appears:** *California Magazine*, Nobel Lecture.
- **Why sourcing is weak:** This is a classic "flashbulb memory" which may be retroactive. Colleagues note he called *many* of his ideas Nobel-worthy.
- **Best evidence:** Retrospective interviews; no contemporary notes from the drive confirm this exclamation.

NARRATIVE LENS RECOMMENDATION

Lens	Score (1-10)	Strengths	Weaknesses
The Rogue Genius	9	Perfect for the "silver Honda" and LSD details.	Risks hagiography; downplays the Cetus team.
The Institutional Rebel	8	Captures the gun threats and CSH "shock" scenes.	Can focus too much on his later controversies.
The Paradigm Shift	10	Best for a BIO 3030 audience; focuses on "Before PCR vs. After PCR."	Can lose the idiosyncratic personal texture.

Final Recommendation: The Paradigm Shift. This lens allows the student to use the "Highway 128" scene as a hook for the moment of conceptual change, while using the "Cetus Conflict" and "DuPont Trial" to show that science is a collective, social, and legal process.

Based on the compiled evidence, here is the assessment of the best narrative lenses and the corresponding evidence cards for a profile of Kary Mullis.

Lens Scoring Rubric

Criterion	Score	Justification

Eureka	3	The Highway 128 drive is a definitively documented "crystallization" moment with specific location, atmospheric, and mathematical details.
Failure	3	The "PCR01" experiment in the purple-capped tube is a documented technical setback that forced the shift from isothermal to thermal cycling.
Mentor	3	Tom White's relationship is pivotal; he protected Mullis's career from Cetus's "interpersonal wrecking ball" consequences twice, acting as his primary institutional shield.
Controversy	3	Multiple high-stakes disputes are well-documented, including the DuPont trial over the Kleppe/Khorana priority and the credit struggle with Erlich and Saiki.
Human Detail	3	Evidence of his rocket-launched frogs, LSD use for scientific visualization, and management of a bakery provides deep character texture.
Last Experiment	2	His late-life project (Altermune) to fight a specific bacterium that killed his friend provides closure but lacks the scene-driven detail of his early work.

Total Score: 17/18

Primary Recommended Lens: The Rogue Genius

Why it fits: This lens accommodates the high-signal "lone inventor" narrative Mullis built around himself while grounding it in the messy, high-friction reality of 1980s biotechnology. It utilizes the best-sourced scenes of conceptual epiphany and laboratory failure to show how a "menace to society" used outsider logic—honed by building backyard rockets and managing a bakery—to revolutionize biochemistry.

- **Best opening scene: SCENE #1: The Night on Highway 128.** This provides the most vivid "hook," combining the sensory details of California buckeyes with the mental calculation of exponential growth ($Y = X \cdot 2^n$).
- **One quote to use early: Quote #3:** "Dear Thor! I exclaimed. With two oligonucleotides, DNA polymerase, and the four nucleoside triphosphates I could make as much of a DNA sequence as I wanted...".
- **Cautions:**
    - **Myth Risk:** The precision of mileage marker 46.58 is likely a retroactive narrative polish; Mullis's own account is the only source for the "Eureka" shouts while his girlfriend slept.
    - **Perspective Bias:** Mullis often minimized the essential validation work done by Randy Saiki and Henry Erlich, which is a key point of dispute among his peers.

Backup Lens: The Institutional Rebel

Score: 16/18 (Justification: Focuses on the friction between individual eccentricity and "Big Science" norms, scoring high on controversy and mentorship but slightly lower on technical closure.)

Why it fits: This lens highlights the institutional drama at Cetus and the legal crucible of the DuPont trial. It portrays Mullis as a disruptor whose breakthrough survived only because of the social and political labor of others (Tom White) and the rigorous validation of "top-notch experimentalists" who initially viewed him as a "wrecking ball".

- **Best opening scene: SCENE #3: The "Interpersonal Wrecking Ball" at Cetus.** This establishes the high-stakes environment where a brilliant idea was nearly smothered by the inventor's own volatile behavior.
- **One quote to use early: Quote #5:** "Science, like nothing else among the institutions of mankind, grows like a weed every year. Art is subject to arbitrary fashion, religion is inwardly focused...".
- **Cautions:** This lens requires careful handling of his late-life controversies (denialism) to avoid overshadowing his genuine scientific contributions.

Final Research Note: BIOL 3030 Bridge

To connect the "Rogue Genius" narrative to the course, focus on the **SCENE #2 (PCR01)** failure. It teaches students that even a "perfect" chemical idea must yield to the thermodynamics of DNA denaturation. Mullis's realization that Bacterial Alkaline Phosphatase (BAP) could be heat-denatured only in zinc-free buffers is a high-signal "gold" detail that connects directly to enzyme regulation and buffer chemistry in intro biochemistry.